

The Memory Cost of Thinking Less: An Examination of Generative AI and Memory Consolidation in the Hippocampus

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Abstract

Use of artificial intelligence is one of the most efficient ways to reduce mental effort (Gerlich, 2025a). With their ability to create full-text summaries and finish work in short periods of time, this substitutive tool risks replacing human thinking through cognitive offloading (Chen et al., 2025; Hashmi et al., 2026). As a result of reduced mental effort, pathways crucial for memory consolidation and reconsolidation are being bypassed, yet the role of generative artificial intelligence in these processes has not yet been established through direct correlation. The study aims to identify the effects of using these tools substitutively in long-term memory consolidation, using both a synthesis of pre-existing literature and analysis of a dataset containing 50,000 post-secondary students. Students were categorized by their primary GenAI use case, and weekly GenAI hours, perceived AI dependency, and skill retention scores were analyzed using statistics such as Pearson correlation coefficients and regression t-tests. The final analysis concluded that use of these technologies substitutively has been associated with a significant decline in long-term retention, though use of the same technology assistively has significantly improved retention. Overall, these findings support the hypothesis that greater reliance on Generative AI may be associated with reduced cognitive engagement and memory retention, while highlighting the need for further longitudinal research using direct neuroimaging measures to examine potential effects on hippocampal memory consolidation.

I. Introduction

Artificial intelligence (AI) is a rapidly evolving tool that has the potential to grow to replace human thinking processes (Gerlich, 2025a). One of the major forms of AI is generative AI (genAI). Due to its nature of formulating responses in short amounts of time and requiring limited cognitive effort – mental effort and energy expended to complete a task – from the user, individuals find themselves using AI tools more often to reduce their mental load (Chen et al., 2025; Hashmi et al., 2026). When we outsource work to AI, whether it be writing essays or solving problems, we free up working memory capacity, thereby reducing cognitive effort and improving efficiency (Lodge & Loble, 2026). This is known as cognitive offloading (Gerlich, 2025a; Lodge & Loble, 2026). AI-induced cognitive offloading could affect our cognitive function alongside our ability to form long-term memories in the hippocampus (Lodge & Loble, 2026). The shift from using assistive technology, such as search engines, to substitutive technology, such as AI, has sparked debate among scientists about how, and whether, AI could affect human cognitive functioning, focusing on memory (Gerlich, 2025a; Lodge & Loble, 2026). Long-term memories form through memory consolidation: a process in which a transient, labile memory becomes more stable and is gradually integrated into long-term memory (Rokshari, 2026). The hippocampus plays an important role in the formation and consolidation of many forms of declarative memory, a type of long-term memory (Dubey et al., 2024; Rokshari, 2026). Through consolidation and reconsolidation, the process by which previously consolidated memories can become temporarily modifiable after retrieval and then stabilized again, our memories are strengthened (Lei et al., 2024). Recall of these memories and this action being repeated can further strengthen memory retention and accessibility (Dubey et al., 2024).

Our aim is to explore the question: **“To what extent does Generative AI-induced Cognitive Offloading Affect Memory Consolidation in the Hippocampus?”** Our research highlights critical information regarding the potential impact of dependency on GenAI on our memory, which is integral for our cognitive functioning and the acquisition and retention of knowledge (Tian & Zhang, 2025). While the process through which GenAI directly affects the brain is still being developed, there have been published studies on its effects on cognitive behaviors and memory-related processes (Chen et al., 2025; Gerlich, 2025b). Using some key tools and techniques used in other studies allow us to examine whether findings on AI use and cognitive offloading may have implications for memory consolidation and hippocampal functioning (Aimé et al., 2015; Kosmyna et al., 2025). These include:

- Electroencephalogram (EEG): a test that measures the electrical activity of the brain, measured by Dynamic Direct Transfer Function (dDTF) (Kosmyna et al., 2025).
- Standardized behavior testing: baseline assessments to measure initial critical thinking, writing skills, and language proficiency (Chen et al., 2025)
- fMRI: non-invasive scanning technique measuring oxygenated blood shifts and cortical hemodynamic activity in the brain (Chen et al., 2025; Dubey et al., 2024)

Existing research has reported lowered cognitive engagement and differences in task performance when individuals rely on AI (Chen et al., 2025; Gerlich, 2025b). However, it

remains unclear whether these findings can be directly connected to memory consolidation and hippocampal processes (Chen et al., 2025; Jose et al., 2025).

II. Literature Review

1. Overview

To examine the relationship between generative AI, cognitive offloading, and memory consolidation, relevant studies were selected based on their connection to at least one of the main themes of our study: cognitive offloading, generative AI, and memory consolidation and the hippocampus. Sources were located through academic databases including PubMed, Google Scholar, and other scholarly databases, and studies were chosen based on their relevance to the research question, methodology, and evidence presented. The literature was then organized by theme to compare findings across different types of research.

2. Cognitive Offloading and Generative AI

A major theme across the literature is that generative AI may replace cognitive tasks that would normally require active mental effort. Jose et al. (2025) describe this as a shift from traditional cognitive aids towards AI acting as a “cognitive surrogate,” in which the tool replaces human mental effort rather than assisting it. They separated cognitive offloading into assistive, substitutive, and disruptive forms, showing that the effects depend on how much of the thinking process is handed over to the technology. Assistive offloading occurs when individuals use tools such as AI to aid memory or data retrieval without replacing internal mental effort, while substitutive offloading occurs when the tool is used to replace cognitive operations. Disruptive offloading occurs when tools are used to completely replace these tasks and is generally associated with the greatest amount of risk of cognitive atrophy, in which neural networks and processes are drastically reduced, and brain function is impaired (Jose et al., 2025). This helps to explain the possible relationship between AI dependence and memory, although the article focuses on opinion rather than direct experimentation.

Gerlich (2025) provides more direct evidence through a mixed-method study using 666 participants. Surveys and interviews were conducted, with quantitative data analyzed through ANOVA and correlation analysis and qualitative responses analyzed thematically. A negative relationship between frequent AI use and critical thinking was found, with cognitive offloading acting as a mediating factor (Gerlich, 2025). While the large sample and mixed methods strengthened the study, its correlational design cannot establish that use of AI directly causes a decline in critical thinking. This is important for our study because it suggests a possible pathway that may also apply to memory, as critical thinking processes data stored in memory.

The study conducted by Gong and Yang (2024) supports the idea that external technology can change how people remember information. Their meta-analysis began with 908 records from five databases and ultimately included 22 articles containing 3 independent comparisons and 30,889 participants. The authors found a moderate, statistically significant overall effect of frequent internet searching on cognitive outcome using Cochrane's Q Test ($Q = 6.1997$, $p <$

0.050). However, the study focused mainly on search engines rather than generative AI, so it provides background for cognitive offloading rather than direct evidence about ChatGPT.

3. Memory Consolidation in the Hippocampus

The connection between cognitive effort and memory consolidation involves stabilizing newly learned information over time. Findings from Gong and Yang (2024) and Sparrow et al. (2011) suggest that when people expect information to remain available externally, they may rely less on their own memory. Their meta-analysis performed by Gong & Yang (2024) also found that the effects varied based on factors such as age, region, and previous internet experience.

Kosmyna et al. (2025) provided some of the most relevant experimental evidence connecting AI use with changes in cognitive activity. Their study divided 54 participants into three separate groups: LLM, search engine, and brain-only. Participants completed essay writing sessions under their respective condition, while EEG was used to measure brain activity. In a later session, some participants switched conditions, known as the swap phase. During this phase, the LLM group had to write essays unaided, using only their brain, and the brain-only group had to write with the aid of LLMs. A major limitation, however, is that not every participant completed the swap phase. Regardless, the group using LLMs primarily showed the weakest neural connectivity, while the brain-only group showed the strongest and most distributed neural networks. In addition, after the swap phase, the LLM group was still unable to recover lost neural activity, causing memory deficits, while the brain-only group experienced heightened neural activity even after the swap phase due to the fact that they used AI as an assistive tool rather than substitutively.

The strength of the study conducted by Kosmyna et al. (2025) is that it used an experimental design and observational EEG data rather than relying only on self-reported AI use. However, the sample was relatively small, and the study measured overall neural activity rather than directly measuring structural changes in the hippocampus. Therefore, it provides evidence of reduced cognitive engagement but does not prove that AI usage can be correlated with structural changes to the hippocampus.

Interactions between the hippocampus and neocortex during learning, memory, and problem solving were examined in the context of the neurological side of our research question using a study by Spens & Burgess (2026). Their research helps explain why hippocampal neocortical interactions matter for memory, but it does not directly test whether generative AI changes the human hippocampus. The significance of this study builds upon current understanding of memory consolidation and the process of rehearsal and retrieval to strengthen neocortical networks and thus consolidate short-term memory into long-term memory.

4. Agreements and Conflicts in the Research

Overall, the studies generally come to a consensus in the fact that externalizing cognitive tasks can reduce the amount of mental effort required from the user, especially when technology provides complete answers rather than simply supporting the learner. However, the disagreement

lies in the use of AI that can be associated with negative effects on memory retention. Jose et al. (2025) emphasize that the effects depend on the type of offloading – assistive, substitutive, disruptive – while other studies suggest that guided or assistive technology can support learning rather than replace it.

Another important difference is the amount of exposure. The existing research does not establish one specific amount of daily or weekly generative AI use that is associated with a memory decline. Instead, studies generally point toward more frequent or dependent use being associated with greater cognitive offloading, though these exact frequencies are not given. For example, stronger cognitive offloading among more frequent AI users was found by Gerlich (2025), while measurable differences between AI-assisted and brain-only conditions were found in a study by Kosmyna et al. (2025), suggesting that both frequency and type of AI usage may matter.

5. Research Gaps

The most significant gap in current research is the lack of direct evidence connecting AI-induced cognitive offloading to structural changes in the hippocampus or the biological process of memory consolidation. Existing studies provide evidence about memory performance, cognitive effort, and brain activity, but they do not clearly show effects in the hippocampus itself as a result of repeated generative AI use over time, in different ways of usage.

6. Hypothesis

Based on the literature, the present study hypothesizes that greater GenAI induced cognitive offloading will be associated with weaker memory performance and reduced memory consolidation, which may indicate reduced engagement of hippocampal processes involved in forming and strengthening long term memories. This does not assume that AI directly damages the hippocampus; rather, it predicts that reduced cognitive engagement and poorer memory performance may be associated with differences in the processes supported by the hippocampus. More direct neuroimaging and longitudinal research would be needed to test the structural part of this hypothesis.

III. Methodology

Our study was conducted in two phases: a literature review of existing peer-reviewed studies and articles on generative AI, cognitive offloading, and hippocampal memory consolidation, and data collection and analysis using an empirical dataset to examine the interrelationship between AI use and cognitive offloading.

We searched for relevant articles using the academic databases: Google Scholar, PubMed, Semantic Scholar, and OpenScience Framework. We included the following keywords to select articles: generative AI, hippocampal memory consolidation, cognitive offloading, AI-induced cognitive offloading, effects of AI on memory. We further filtered out articles published over the past 10 years. Articles were selected based on their ability to meet the following criteria in the respective order: passing the review, being published under a reputable journal (including but not limited to Nature, the American Association for the Advancement of Science, and Neuron),

being peer-reviewed, and direct connection with memory consolidation, cognitive unloading or hippocampal function; priority was given to articles with the availability of empirical data. After selecting the articles and checking their accuracy, reliability, and relevance for our research task, we grouped them into a table of evidence to summarize key findings from each study, including the methodology used, results found in the study, and remaining gaps in understanding. We then compared results from all studies using a theme-based analysis to identify similarities, differences, conflicts between articles, and trends across the core themes: cognitive offloading and generative AI, memory consolidation and the hippocampus, the effects of Generative AI on learning, memory, and brain function, and research gaps and future directions.

We searched for datasets from the following databases: Kaggle Datasets Repository, Google Dataset Search, GitHub, ICSPR, and OpenNeuro. Datasets were chosen based on their key variables and relevance to the question. We focused on using one dataset examining the impacts of Generative AI on students in post-secondary. This dataset contained values for 50,000 students, where we examined the key variables: Primary Use Case, Weekly GenAI Hours, Perceived AI Dependency, and Skill Retention Score. We started by cleaning the raw data by deleting all duplicate values or strings, as they can distort the results; removing empty cells with missing values; data standardization (single format, text type), correction of errors (typos, extra spaces); and removal of emissions. We calculated the average value and standard deviation of each group to assess the central trend and the spread of GenAI usage and skill retention among participants. Statistical analysis was carried out using both Google Sheets and Microsoft Excel to determine significant differences among the groups using GenAI for the different purposes.

IV. Results

We analyzed a dataset compiled by Sisiro (2026) about the impact of AI on students. This dataset featured 50,000 post-secondary students across freshman, sophomore, junior, senior year, or graduate. We cleaned the data to include only relevant variables: student ID, primary use case, weekly GenAI hours, perceived AI dependency, and skill retention score. Student ID was a randomized number assigned to each student. GenAI hours were the average weekly hours the student spent using GenAI. Perceived AI dependency was a student-based rank on a scale from 1(low) to 10(high) on their own dependency on AI tools. Skill retention score is a score (from 0-100) measuring how well students retained learned skills over the semester. Using Microsoft Excel and Google Sheets, we were able to sort all students into five groups based on their primary use case of GenAI: copywriting/drafting (C/D), debugging/troubleshooting (D/T), direct answer generation (DAG), ideation (I), and summarizing reading (SR). From these groups, we determined the total and group mean and population standard deviation weekly GenAI hours, perceived AI dependency, and skill retention score. Our results can be found in Table 1 below. Note that we used Microsoft Excel to perform mean and population standard deviation calculations. Note that all raw data values were obtained from the Sisiro (2026) dataset, accessed August 13, 2026, however all analysis has been done by us.

Table 1: Mean (μ) and Population Standard Deviation (σ) of: Weekly GenAI Hours, Perceived AI Dependency, and Skill Retention Score

Category	Analysis	Total	C/D	D/T	DAG	I	SR
Weekly GenAI Hours	μ (± 0.001 hrs)	8.428	7.628	9.401	8.456	8.398	8.171
	σ (± 0.001 hrs)	8.269	7.500	9.075	8.389	8.164	7.988
Perceived AI Dependency	μ (± 0.1)	3.5	3.4	3.6	3.5	3.5	3.5
	σ (± 0.1)	1.8	1.7	1.9	1.8	1.8	1.8
Skill Retention Score	μ (± 0.001)	75.798	75.231	78.069	73.719	75.520	75.225
	σ (± 0.001)	13.281	12.722	13.900	14.300	12.641	12.699

Mean sample calculation (Total weekly GenAI Hours)

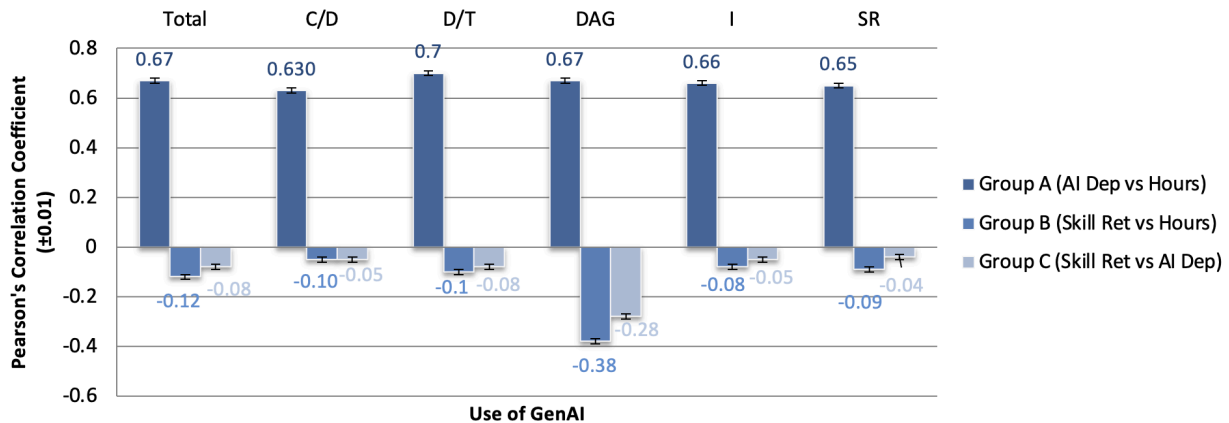
$$\mu = \frac{\sum x}{N} = \frac{(23.31)+(1.82)+\dots+(15.87)}{50000} = 8.428 \text{ hours}$$

Standard deviation sample calculation (Total weekly GenAI Hours)

$$\sigma = \sqrt{\frac{\sum (x-\mu)^2}{N}} = \sqrt{\frac{(23.31-8.428)^2+(1.82-8.428)^2+\dots+(15.87-8.428)^2}{50000}} = 8.269 \text{ hours}$$

Next, we decided to calculate the regression on various variables to assess for any correlation between variables. We calculated regression between perceived AI dependency vs. weekly GenAI hours (A), skill retention score vs. weekly GenAI hours (B), and skill retention score vs. perceived AI dependency (C) for each group and also the overall total. These results can be summarized in Figure 1 below. Note that this was also done utilizing Microsoft Excel.

Figure 1: Regression values calculated using Pearson's Correlation Coefficient for all groups and total overall



Pearson's Correlation Coefficient Sample Calculation (Group A, Total)

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}}$$

$$r = \frac{50000[(23.31)(5) + \dots + (15.87)(4)] - [(23.31) + \dots + (15.87)][(5) + \dots + (4)]}{\sqrt{[(50000)[(23.31)^2 + \dots + (15.87)^2] - (23.31 + \dots + 15.87)^2][(50000)[(5)^2 + \dots + (4)^2] - (5 + \dots + 4)^2}} = 0.67$$

After, in order to analyze and test for significant differences from each group to the total, we performed a regressional t-test for each group. In the t-test, there are two hypotheses. The null hypothesis (H_0) states that the Pearson's correlation coefficient obtained from the group is not significant from the value obtained for the group, and any differences are due to chance. The alternative hypothesis (H_1) states that the Pearson's correlation coefficient obtained from the group is significant from the value obtained for the group, and thus differences are not due to chance. All the results for this can be found in Table 2 below. Note that these were performed with unrounded values. The regression t-test was done by hand for each group using a Graphical Display Calculator (GDC).

Table 2: Regression t-test values calculated for each group, compared to the total. Note that a critical value of ± 1.960 was used for all groups due to the relatively large sample size and a p-value of $p=0.05$ was used to account for a slight margin that results could be due to chance.

Regression Group (A, B, C)	Category Compared to Total	T-statistic (± 0.01)	Degrees of Freedom (df)	Hypothesis Accepted
A	C/D	-5.79	12009	H_1
	D/T	+6.25	12293	H_1
	DAG	+0.86	6339	H_0
	I	-0.53	10719	H_0
	SR	-2.35	8631	H_1
B	C/D	+8.05	12009	H_1
	D/T	+1.76	12293	H_0
	DAG	-24.83	6339	H_1
	I	+4.17	10719	H_1

	SR	+3.34	8631	H_1
Regression Group (A, B, C)	Category Compared to Total	T-statistic (± 0.01)	Degrees of Freedom (df)	Hypothesis Accepted
C	C/D	+4.50	12009	H_1
	D/T	+0.47	12293	H_0
	DAG	-17.62	6339	H_1
	I	+3.82	10719	H_1
	SR	+4.31	8631	H_1

T-Test Sample Calculation (Group A, Category C/D, against Total)

H_0 : regressions are not significantly different and any changes are caused by chance

H_1 : regressions are significantly different and any changes are not caused by chance

$$t = \frac{b_{group} - b_{total}}{\sqrt{SE_{b_{group}}^2 - SE_{b_{total}}^2}} = \frac{(0.6292451795153072) - (0.6654785561440112)}{\sqrt{(0.0070922364)^2 - (0.0033381449)^2}} = -5.79^1$$

$$df = n_{group} - k = 12011 - 212009^2$$

Since $-5.79 < \pm 1.60$, then we reject the null hypothesis and accept the alternate hypothesis; this group's data was significantly significant from the total.

V. Discussion

To evaluate how digital cognitive offloading impacts human memory mechanisms, we synthesized empirical data points from neuroimaging studies, a meta-analytical review on internet search behavior, and quantitative surveys on artificial intelligence tools. We also compared these to the dataset we analyzed from Sisiro et al. (2025). The compiled data documents a progressive shift from assistive search engines to substitutive Generative AI, illustrating how modern tools alter memory consolidation and critical thinking (Jose et al., 2025; Kosmyna et al., 2025).

Brain imaging data shows that moving from assistive search to substitutive AI causes a significant decrease in active neural processing. Brain scans using EEGs, fMRIs, and Dynamic Direct Transfer modeling from Kosmyna et al. (2025) show that relying on Generative AI reduces neural activity in the prefrontal cortex and hippocampus much more compared to when

¹ Note that b (slope) can be representative of the calculated r-statistic. SE is standard error from mean.

² Note that n is the sample size in the subgroup, k is the number of estimated coefficients.

traditional search engines are used. EEG and fMRI data showcased scenes that rely on Generative AI to generate answers, resulting in a 55% reduction in neural activity across the prefrontal and hippocampal networks compared to unassisted problem solving. By contrast, using web search engines to locally source material led to a milder 34-48% drop (Kosmyna et al., 2025). To form long-term memories, the brain needs strong, repeated electrical signals across the prefrontal hippocampal networks to trigger synaptic activity and build lasting neural pathways (Kosmyna et al., 2025; Spens & Burgess, 2026). When AI automates the thinking process, the brain lacks the active and cognitive stimulus required to fire these circuit loops. Without this necessary stimulus, synaptic activity stays too weak, leaving the hippocampus incapable of encoding and storing new information into long-term memory (Gerlich, 2025a; Spens & Burgess, 2026). Dynamic Directed Transfer modeling demonstrates that automating higher-order tasks deprives the prefrontal executive networks and hippocampus memory circuits of the active firing loops needed to build and sustain lasting neural pathways. Over time, this creates cognitive debt where repeated use of AI and lack of stimulus to the hippocampal area in the brain leads to cumulative loss in memory retention and independent problem-solving skills.

The heavy reliance on search engines creates substantive learning where the human brain adapts to remember where to find information rather than the information itself. Meta-analytic data shows that offloading search to external tools significantly changed cognitive load and reduced memory encoding in the hippocampus. A meta-analysis done on the Google effect (Gong & Yang, 2024) showcases that search engine reliance creates a transactive memory system using Cochran's Test. Specifically, regular search engine usage significantly increased reliance on external memory tools ($Q = 6.1997$, $p < 0.050$), while fast online retrieval created an overestimation of personal knowledge. This is because users confuse interactive access with true internal understanding, creating an "illusion of competence" (Jose et al., 2025). Additionally, search engine reliance also significantly altered everyday information seeking behaviors. While this form of offloading reduces cognitive load during retrieval, it fundamentally alters memory encoding in the hippocampus by training the brain to track only external links.

As offloading moves from search engines to Generative AI, the delegation shifts from external memory storage to delegating executive reasoning. Empirical analysis done by Gerlich (2025) on AI reliance reveals that cognitive offloading acts as a direct mediating mechanism through which AI tool usage reduces critical thinking and reflective judgement. Quantitative analysis demonstrating a strong and statistically significant negative correlation between cognitive offloading to AI tools and critical thinking scores ($r = -0.75$, $p < 0.001$), with a multiple regression model establishing that AI tool usage significantly predicts lower critical thinking scores. This breakdown occurs because critical thinking relies on working memory and deliberate mental effort to analyze complex arguments and identify flaws (Gerlich, 2025a; Tian & Zhang, 2025). Additionally, a formal mediation analysis revealed that cognitive offloading significantly mediates the negative impact of AI usage on critical thinking, with the indirect effect through offloading accounting for a majority of this group and remaining is done by direct effect. This higher user trust on automated AI systems encourages cognitive offloading, causing

the prefrontal executive networks to bypass active working memory processing required to form long-term memories in the hippocampus, leaving the active neural loops needed for complex thinking completely unused.

The dataset we analyzed from Sisiro (2026) shows similar findings. We found a lower mean skill retention score in those using AI for direct answer generation than other groups using AI for other purposes. It is worthy to note, however, that this group also had the highest standard deviation, indicating the data had a greater spread around the mean. We also found a total moderate positive correlation between perceived AI dependency and weekly GenAI hours, supporting our hypothesis that increased GenAI usage increases dependence, and weak negative correlation between skill retention score and weekly GenAI hours. Skill retention score was not found to be correlated with perceived AI dependency. Although the dataset measured skill retention rather than memory retention, it is important to note that skill retention is a subset of memory retention, and is more specifically procedural memory. The negative correlation found supports our hypothesis that increased use of GenAI reduces memory retention, however, the fact that this correlation was very weak indicates possible discrepancies between how one uses AI and its effect on skill retention. The regression t-test performed between each group and the overall total highlighted significant differences in weekly GenAI hours and skill retention scores, as well as in perceived AI dependency and skill retention scores between both groups using GenAI for copywriting/drafting and direct answer generation when compared to the total. For the group using GenAI for direct answer generation, a drastic test statistic was found between both weekly GenAI hours and skill retention score ($t = -24.83$), as well as between perceived AI dependency and skill retention score ($t = -17.62$) indicating a significantly stronger negative correlation between the two variables, respectively. Another important thing to note is that the regression values for perceived AI dependency and skill retention score display a weak negative correlation as opposed to no correlation found in all other groups, therefore indicating that those perceiving increased dependency on GenAI for direct answer generation experience lower skill retention score, supporting our hypothesis that increased dependency on GenAI is associated with decreased memory retention. As for those using GenAI for copywriting/drafting, the largest positive test statistic was seen between both weekly GenAI hours and skill retention score ($t = +8.05$), as well as between perceived AI dependency and skill retention score ($t = +4.50$) indicating a significantly weaker correlation ($r = -0.05$ for both) for those using GenAI for copywriting/drafting when compared to the total. This demonstrates that the matter GenAI is used for is significant when determining if the correlation between GenAI use is positive or negative, in respect to skill retention.

We decided to not remove outliers in the dataset analyzed due to them having minimal effect on calculations as a result of the large sample size (50,000 students).

A major strength of this study is its multi-layered approach is its ability to combine empirical evidence from peer-reviewed meta-analyses with direct brain imaging scans and statistical models to provide a full picture of cognitive offloading. The dataset analyzed also

contained a large sample size (50,000 students), therefore results are more likely to reflect the general population.

However, a few general limitations in the current research must be kept in mind. First, because Generative AI is so new, there is not enough long-term data showing how years of AI use affect brain structure and learning habits over time. Many studies constantly address this issue saying more long-term experimentation is required to accurately assess these effects. Second, much of the behavioral research relies on correlational data, which shows a strong link between AI offloading and lower thinking performance but cannot prove that AI directly causes the drop. Third, the dataset from Sisiro (2026) was analyzed with perceived AI dependency: a subjective value ranging from 1-10. Subjective ratings may not reflect objective dependency of GenAI usage and interpretation of in-between values are subject to change between individuals. Fourth, data used from this dataset is specific to procedural long-term memory, rather than all types of long-term memory retention. Fifth, data collected from these studies were not inclusive of the entire population. The dataset analyzed from Sisiro (2026), although containing 50,000 responses, only contained the student population in post-secondary, or recent graduates. Unequal sample sizes were also found between those who used GenAI for Direct Answer Generation ($n = 6340$) while the rest of the four groups usually had over 10,000 responses, thus leading to results skewed towards certain groups especially when calculating totals. In addition, the relatively small sample size ($n = 54$) of the study conducted by Kosmyna et al. (2025) does not accurately represent the whole population, especially due to the fact that the majority of participants were ages 18-39 and from Boston-area colleges. More experimentation will need to be done especially in the age categories of adolescents and children, as well as older adults and seniors.

Overall, the synthesized data reveals a fundamental shift in how humans offload cognitive tasks. Older digital tools delegated memory storage; modern AI tools delegate executive reasoning. This transition addresses the central research question by demonstrating that as offloading moves from finding information to automated thinking, the human brain is progressively bypassed during critical stages of processing. This progression carries immense real-world importance, as it highlights how Generative AI can directly alter brain-based learning techniques.

VI. Conclusion

Therefore, these findings indicate that generative AI-induced cognitive offloading influences the process of memory consolidation in the hippocampus. The use of Large Language Models, specifically Generative AI, has been correlated with the outsourcing of the retrieval practice to strengthen those memory neural pathways (Ali et al., 2024). However, a critical point that must be highlighted is that this only occurs when AI is used substitutively or disruptively (Jose et. al, 2025). In situations where AI is used for specific tasks, such as a scaffolding tool to support active recall and provide personalised feedback, it has been shown to cause improved memory retention both short- and long-term (Rokshari, 2026). This strengthens our initial

hypothesis as these activities still sustain cognitive engagement without supplanting executive function, thus leading to improved retention of memory (Hashmi et al., 2026).

The significance of this study is to find the “limit” of productive AI usage before it negatively correlates to reduced memory and retention, and to analyse its general effects on cognitive function so the general population can use it to optimize both cognitive efficiency and long-term retention whilst minimizing any possible negative impacts. We believe that these technologies can indeed be helpful with the inclusion of certain constraints/some guardrails, including but not limited to

- Momentarily terminating AI input to force users to actively engage with their output (Jose et. al, 2025)
- Requiring users to rate their confidence levels with the concept to prevent the “illusion of competence” (Gerlich, 2025b)
- Quizzing users about the mechanisms of the ideas being discussed to prevent the “Google Effect” (Gong & Yang, 2024; Sparrow et. al, 2011)
- Designing systems that regulate the pace of information being presented to ensure the user can actively engage with the ideas (Dubey et. al, 2024)

It should be noted that there are some limitations within our methodology, including the limited number of studies with empirical data and analysis, and studies regarding the neurological structure, specifically the hippocampus, through the use of technologies such as MRI, DTI, etc.; it should also be highlighted that the majority of our data is based on correlational influences across studies and that there is a lack of long term experimentation available within the field. Future directions would include collecting more longitudinal direct evidence using MRI and DTI technologies to measure structural changes.

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